Complex Analysis – new problems

Problems built on recent research results

Each problem is a special case of a published result or of a lemma used in its proof; the papers are in this folder. The problems are ordered by difficulty. The difficulty is the position of the problem in an IMC day: 1 easy, 2 below average, 3 medium, 4 above average, 5 hard.

Problem 1. *IMC level: 1/5 (easy)*

Let a, b be points of the unit disc $\mathbb{D}$ and

$$B(z) = \frac{z-a}{1-\bar{a}z} \cdot \frac{z-b}{1-\bar{b}z}.$$

Prove that B' has exactly one zero c in $\mathbb{D}$, and that there is a conformal automorphism φ of $\mathbb{D}$ with $\varphi(c) = 0$ and $\varphi(a) = -\varphi(b)$.

Based on: A. N. Fletcher, A. Hill, *On cubic Blaschke products*, arXiv:2511.13484, Section 1.2.

Problem 2. *IMC level: 2/5 (below average)*

Let $u\colon \mathbb{D} \to (0,\infty)$ be a harmonic function on the unit disc with $u(0) = 1$, and put $c = \frac{1}{2}|\nabla u(0)|$. Prove that $c \leqslant 1$ and that for every $z \in \mathbb{D}$, $r = |z|$,

$$\frac{1-r^2}{1+2cr+r^2} \leqslant u(z) \leqslant \frac{1+2cr+r^2}{1-r^2}.$$

Show that for every $c \in [0,1]$ and $r \in (0,1)$ each of the two inequalities can turn into an equality.

Based on: M. Svetlik, *A strengthening of the Harnack inequality*, arXiv:2501.09880, Theorem 1.

Problem 3. *IMC level: 2/5 (below average)*

Let $n > 0$.

(a) Prove that for every $s \in (-1,1)$

$$\int_0^\infty \frac{x^s \cos\left(nx - \frac{\pi s}{2}\right)}{1+x^2}\,dx = \frac{\pi}{2}e^{-n}.$$

(b) Deduce the identities

$$\int_0^\infty \frac{\cos(nx)\log x}{1+x^2}\,dx = -\frac{\pi}{2}\int_0^\infty \frac{\sin(nx)}{1+x^2}\,dx,$$

$$\pi\int_0^\infty \frac{\sin(nx)\log x}{1+x^2}\,dx + \int_0^\infty \frac{\cos(nx)\log^2 x}{1+x^2}\,dx = \frac{\pi^3}{8}e^{-n}.$$

Based on: B. C. Berndt, A. Straub, *Certain integrals arising from Ramanujan's notebooks*, arXiv:1509.00886, Theorems 2.1 and 3.1.

Problem 4. *IMC level: 3/5 (medium)*

Let $f\colon \mathbb{D} \to \mathbb{D}$ be holomorphic with $f'(0) = 0$. Prove that

$$\frac{|f'(z)|}{1-|f(z)|^2} \leqslant \frac{2|z|}{1-|z|^4} \qquad (z \in \mathbb{D}),$$

and find all f for which equality holds at some point $z \neq 0$.

Based on: A. Moucha, *A Burns–Krantz type theorem for Blaschke products*, arXiv:2505.21346, Theorem E.

Problem 5. *IMC level: 3/5 (medium)*

Find all pairs (c, f), where $c \in \mathbb{C}$ and f is an entire function satisfying $|f(z)| \leqslant Ae^{A|z|}$ for some constant $A > 0$, such that

$$f'(z)^2 + f(z+c)^2 = 1 \qquad \text{for all } z \in \mathbb{C}.$$

Based on: G. Haldar, *On entire solutions of system of Fermat-type difference and partial differential-difference equations in* $\mathbb{C}^n$, arXiv:2310.20358, Introduction.

Problem 6. *IMC level: 3/5 (medium)*

For a complex number q with $0 < |q| \leqslant \frac{1}{5}$ let

$$\theta(q, x) = \sum_{j=0}^{\infty} q^{j(j+1)/2} x^j \qquad (x \in \mathbb{C}),$$

an entire function of x.

(a) Prove that $\theta(q, \cdot)$ has no zeros in the disc $|x| \leqslant |q|^{-1/2}$ and exactly one zero in each annulus $|q|^{-k+1/2} < |x| < |q|^{-k-1/2}$, $k = 1, 2, \ldots$; in particular, all its zeros are simple.

(b) Prove that for $0 < q \leqslant \frac{1}{5}$ all zeros of $\theta(q, \cdot)$ are real and negative.

Based on: Y. Gati, V. P. Kostov, *Where not to find the spectrum of the partial theta function*, arXiv:2605.29903, Theorem 1.

Problem 7. *IMC level: 4/5 (above average)*

Let f be holomorphic in a neighbourhood of the closed unit disc, with $f(\mathbb{D}) \subset \mathbb{D}$ and $f(0) = f'(0) = 0$. Let $\sigma_1, \ldots, \sigma_n$ be distinct points of the unit circle with $f(\sigma_j) = \sigma_j$. Prove that each $f'(\sigma_j)$ is a real number greater than 1 and that

$$\sum_{j=1}^{n} \frac{1}{f'(\sigma_j) - 1} \leqslant 1.$$

Based on: M. Abate, *Multipoint Julia theorems*, arXiv:2101.03559, Theorem 1.8 and Proposition 3.11.

Problem 8. *IMC level: 4/5 (above average)*

For $w \in \mathbb{D}$ let

$$B_w(z) = \Big(\frac{z-w}{1-\bar{w}z}\Big)^2.$$

Prove that B_w has a fixed point in the open unit disc if and only if w lies inside the cardioid $\{\frac{1}{3}(e^{2i\theta} - 2e^{i\theta}) : \theta \in [0, 2\pi)\}$, that is, if and only if $w = -\frac{1}{3} + \rho e^{i\phi}$ with $0 < \rho < \frac{2}{3}(1 + \cos\phi)$.

Based on: A. N. Fletcher, A. Hill, *On cubic Blaschke products*, arXiv:2511.13484, Section 4.1.

Problem 9. *IMC level: 4/5 (above average)*

(a) Prove that for every positive integer n

$$\int_0^\infty \frac{x^3 \sin(nx)}{\cosh x - \cos x}\, dx = 0.$$

(b) Prove that for every odd positive integer n

$$\int_0^\infty \frac{\sin(nx)}{x(\cosh x + \cos x)}\, dx = \frac{\pi}{4}.$$

Based on: C. Xu, J. Zhao, *General Berndt-type integrals and series associated with Jacobi elliptic functions*, arXiv:2401.01385, Introduction; X. Chen, C. Xu, J. Zhou, arXiv:2510.16041, Introduction.

Problem 10. *IMC level: 5/5 (hard)*

(a) Let $f\colon \mathbb{D} \to \mathbb{D}$ be holomorphic and suppose that $f(r) = r + o\big((1-r)^3\big)$ as $r \to 1^-$, $r \in (0,1)$. Prove that $f(z) = z$ for all $z \in \mathbb{D}$.

(b) Show that the function $B(z) = \dfrac{3z^2+1}{z^2+3}$ maps $\mathbb{D}$ into $\mathbb{D}$ and that $B(r) - r \sim \frac{1}{4}(1-r)^3$ as $r \to 1^-$; thus the exponent 3 in (a) cannot be replaced by a smaller one.

Based on: A. Moucha, *A Burns–Krantz type theorem for Blaschke products*, arXiv:2505.21346, Theorems A and B; F. Bracci, D. Kraus, O. Roth, arXiv:2310.05521, Remark 3.